



Bits and Atoms

The \$400 Limit Applies To Atoms Only

When returning from abroad, you must complete a Customs declaration form. But have you ever declared the value of the bits you acquired while travelling? To Customs officers, the value of any diskette is the same—full or empty—only a few dollars; the value of the atoms.

I recently visited the headquarters of one of the top five integrated-circuit manufacturers in the US. I was asked to sign in and, in the process, was asked whether I had a laptop with me. I did. The receptionist asked for the model, serial number, and the computer's value. "Roughly US\$1 to \$2 million," I said. "Oh, that cannot be, sir," she replied. "What do you mean? Let me see it."

I showed her my old PowerBook, and she estimated its value at \$2,000. She wrote down that amount and I was allowed to enter.

Our mindset about value is driven by atoms. The General Agreement on Tariffs and Trade is about atoms. New movies and music are shipped as atoms. Companies declare their atoms on a balance sheet and depreciate them according to rigorous schedules. But their bits, often far more valuable, do not appear. Strange.

because it commands such a small piece of the market. By contrast, Blockbuster can pull Hollywood around by the nose, because video stores provide 50 per cent of Hollywood's revenues and 60 per cent of its profits.

I thought about Huizenga's remark and realised that this extraordinary entrepreneur did not understand the difference between bits and atoms. His atoms—videocassettes—prove that video-on-demand will work. Videocassettes are pay-per-view TV. The only difference is that in his business he can draw as much as one-third of the profits from late fees.

Library Of The Future

Thomas Jefferson introduced public libraries as a fundamental American right. What he never considered was that every citizen could enter every library and borrow every book simultaneously, with a keystroke, not a hike. All of a sudden, those library atoms become library bits and are accessible to anyone on the Net. This is not what Jefferson imagined. This is not what authors imagine. Worst of all, this is not what publishers imagine.

The problem is simple. When information is embodied in atoms, there is a need for all sorts of industrial-age means and huge corporations for delivery. But

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Atoms Are Judged Less Greene Than Bits

When Judge Harold Greene broke up AT&T in 1983, he told the newly created regional Bell operating companies that they could not be in the information business. Who did he think he was fooling? The seven sisters were already in the information business and doing just fine, thank you. Their largest margins were—and still are—from the Yellow Pages, which they have sold at great profit. Judge Greene, sir, the companies are and always have been in the information industry. What are you talking about?

What the judge is saying is that the companies have every right to kill thousands of trees, to litter our homes, and to fill garbage sites with their information business, as long as this information is in the form of atoms. But as soon as the companies deliver the exact same information with no-deposit, no-return, environmentally friendly bits, they have broken the law.

Doesn't that sound screwy? Was anyone thinking about the meaning of 'being digital' during the time that AT&T was being disassembled? I fear not.

Pay Per View

During a speech I gave at a recent meeting of shopping centre owners, I tried to explain that a company's move into the digital future would be at a speed proportionate to the conversion of its atoms to bits. I used videocassette rental as an example, since these atoms could become bits very easily.

Wayne Huizenga, Blockbuster's former chairman, was the lunch speaker. He defended his stock by saying, "Professor Negroponte is wrong." His argument was based on the fact that pay-per-view TV has not worked

suddenly, when the focus shifts to bits, the traditional big guys are no longer needed. Do-it-yourself publishing on the Internet makes sense. It does not for paper copy.

Markoff-On-Production

It was through *The New York Times* that I came to know and enjoy the writing of computer and communications business reporter John Markoff. Without *The New York Times*, I probably would not have been introduced to him. However, now it would be far easier for me to collect his new stories automatically and drop them into my personal newspaper or suggested reading file. I would be willing to pay Markoff five cents for each of his new pieces.

If one-fiftieth of the 1995 Internet population subscribed to this idea, and Markoff wrote 20 stories a year, he would earn \$1 million, which I am prepared to guess is more than *The New York Times* pays him. If you think one-fiftieth is too large a percentage, then wait awhile. Once someone is established, the added value of a distributor becomes less and less in a digital world.

The distribution and movement of bits is much easier than atoms. But delivery is only part of the issue. A media company is, among other things, a talent scout, and its distribution channels, bits or atoms, provide a testbed for public opinion. But after a certain point, the author may not need this forum. In the digital age, *Wired* authors can sell their stories direct and make more money, once they are discovered.

While this does not work today, it will work very well, very soon—when 'being digital' becomes the norm. ■

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